

Spatial Age

Five Vision Pro minigames, played with a bare fingertip in the user's own room, produce 18 measurements of speed, decision, control, memory and consistency. ScoreKit turns them into one functional age with an uncertainty interval.

23

METRICS EXTRACTED

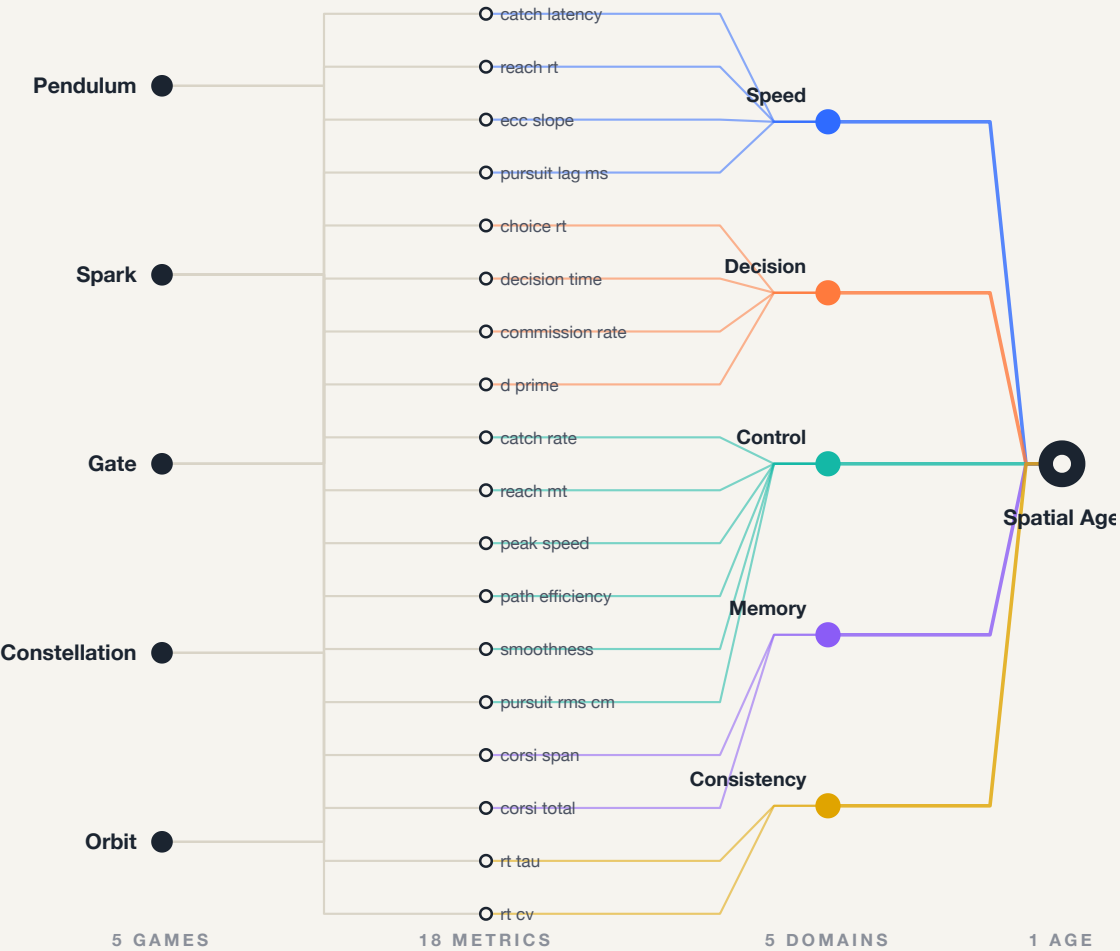
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DOMAINS FUSED

6 min

SESSION

Every number in this document is produced by ScoreKit 0.2.0 with norm table prior-0.2 on synthetic sessions from ScoreKitSynth. No participant data yet.

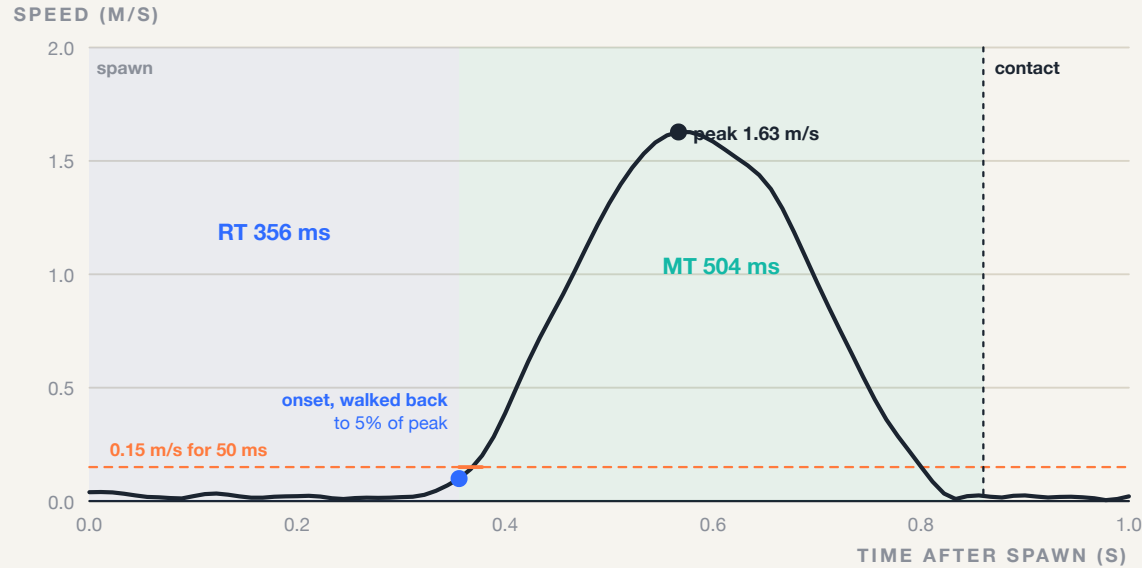


Five minigames, one session file

Each game opens with unscored familiarization trials. Every scored trial logs a 90 Hz fingertip trace, timestamps and a per-trial hand tracking gap. Reach trials with a gap over 100 ms are dropped; Orbit masks gaps up to 300 ms.

01 Pendulum Catch the weight when the cord lets go. TRIALS 3 + 12 MEASURES Catch latency, drop distance, catch rate 28.3cm Drop distance, SYN024	02 Spark Touch each light as it appears. TRIALS 5 + 20 MEASURES Reaction, movement, reach kinematics 349ms Reaction time, SYN024	03 Gate Touch blue. Leave orange. TRIALS 5 + 30 MEASURES Decision time, inhibition, d' 194ms Decision time, SYN024	04 Constellation Watch the stars light, then touch them in order. TRIALS 1 + 14 MEASURES Spatial working memory span 5.0 Corsi span, SYN024	05 Orbit Keep your fingertip inside the moving light. TRIALS 1 + 3 MEASURES Tracking error, lag, velocity gain 5.8cm Tracking error, SYN024
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From fingertip trace to reaction and movement time



Synthetic minimum-jerk reach run through the same pipeline as `Kinematics.swift`.

- 1 **Resample**
Linear interpolation onto a uniform 90 Hz grid.
- 2 **Smooth**
Gaussian kernel, $\sigma = 2$ samples (22 ms).
- 3 **Differentiate**
Central difference velocity, $\text{speed} = |v|$.
- 4 **Onset**
First run of speed above 0.15 m/s held for 50 ms, then walked back to where speed last fell below 5% of peak. Device `move_t` is a fallback.
- 5 **Gate**
RT under 100 ms is an anticipation, over 1500 ms a lapse. Both drop from timing stats.
- 6 **Reach shape**
Peak speed, path efficiency, log dimensionless jerk (LDLJ) and submovement count between onset and contact.

23 metrics, each with a standard error

Medians are taken after dropping values more than 3 scaled MADs from the median. The standard error feeds the age model, so a game played with few clean trials carries less weight. τ is the norm's residual error as an age predictor, in years.

GAME	METRIC	ESTIMATOR	SE	DOMAIN	τ (y)
Pendulum	catch_latency	Robust median of grasp time minus release; < 100 ms dropped as anticipation	$1.2533 \cdot s / \sqrt{n}$	● speed	14
	catch_drop_cm	Robust median of fall distance at grasp; ruler-drop $d = g \cdot t^2 / 2$	display	display	
	catch_rate	Catches / attempts	Agresti-Coull	● control	26
Spark	reach_rt	Robust median, trace onset minus spawn, 100 to 1500 ms kept	$1.2533 \cdot s / \sqrt{n}$	● speed	13
	reach_mt	Robust median, contact minus onset	$1.2533 \cdot s / \sqrt{n}$	● control	16
	rt_tau	Ex-Gaussian τ by method of moments, untrimmed RTs, $n \geq 8$	default	● consistency	18
	rt_cv	SD / mean of MAD-trimmed RTs	default	● consistency	18
	ecc_slope	OLS of RT + MT on eccentricity / 90°	OLS slope SE	● speed	22
	peak_speed	Robust median of per-reach peak speed	$1.2533 \cdot s / \sqrt{n}$	● control	20
	path_efficiency	Straight distance / path length	$1.2533 \cdot s / \sqrt{n}$	● control	22
	smoothness	$LDLJ = -\ln(D^3 / v^2 \text{peak} \cdot \int jerk ^2 dt)$	$1.2533 \cdot s / \sqrt{n}$	● control	20
Gate	choice_rt	Robust median RT on Gate hits	$1.2533 \cdot s / \sqrt{n}$	● decision	13
	decision_time	Choice RT minus Spark RT	$\sqrt{(SE_1^2 + SE_2^2)}$	● decision	18
	commission_rate	False alarms / kept trials	Agresti-Coull	● decision	26
	omission_rate	Misses / go trials	flag only	display	
	d_prime	$\Phi^{-1}(H) - \Phi^{-1}(F)$, log-linear correction (Hautus 1995)	default	● decision	22
Constellation	corsi_span	Longest sequence recalled in order	fixed 0.5	● memory	16
	corsi_total	Span × correct sequences (Kessels 2008)	default	● memory	17
	corsi_tap_interval	Robust median gap between recall taps	display	display	
Orbit	pursuit_rms_cm	RMS fingertip-to-target distance after 1 s acquisition	$SD / \sqrt{\text{trials}}$	● control	16
	pursuit_lag_ms	Grid search 0 to 500 ms, 5 ms step, min summed distance	$SD / \sqrt{\text{trials}}$	● speed	18
	pursuit_on_target	Share of samples within 4 cm	display	display	
	pursuit_gain	Fingertip velocity projected on lag-aligned target velocity	display	display	

Each metric is inverted through an age curve



Reach RT for SYN024 against the provisional norm (Der and Deary 2006 shape).

$$E[x \mid a] = \mu_{25} + \beta (a - 25) + \gamma \cdot \max(0, a - 50)^2$$

Linear decline from 25 with an extra quadratic term after 50, the shape reported for processing speed and Corsi span.

$$\hat{a} = E^{-1}(x)$$

Bisection on the monotone curve, 60 iterations, clamped to 18 to 95.

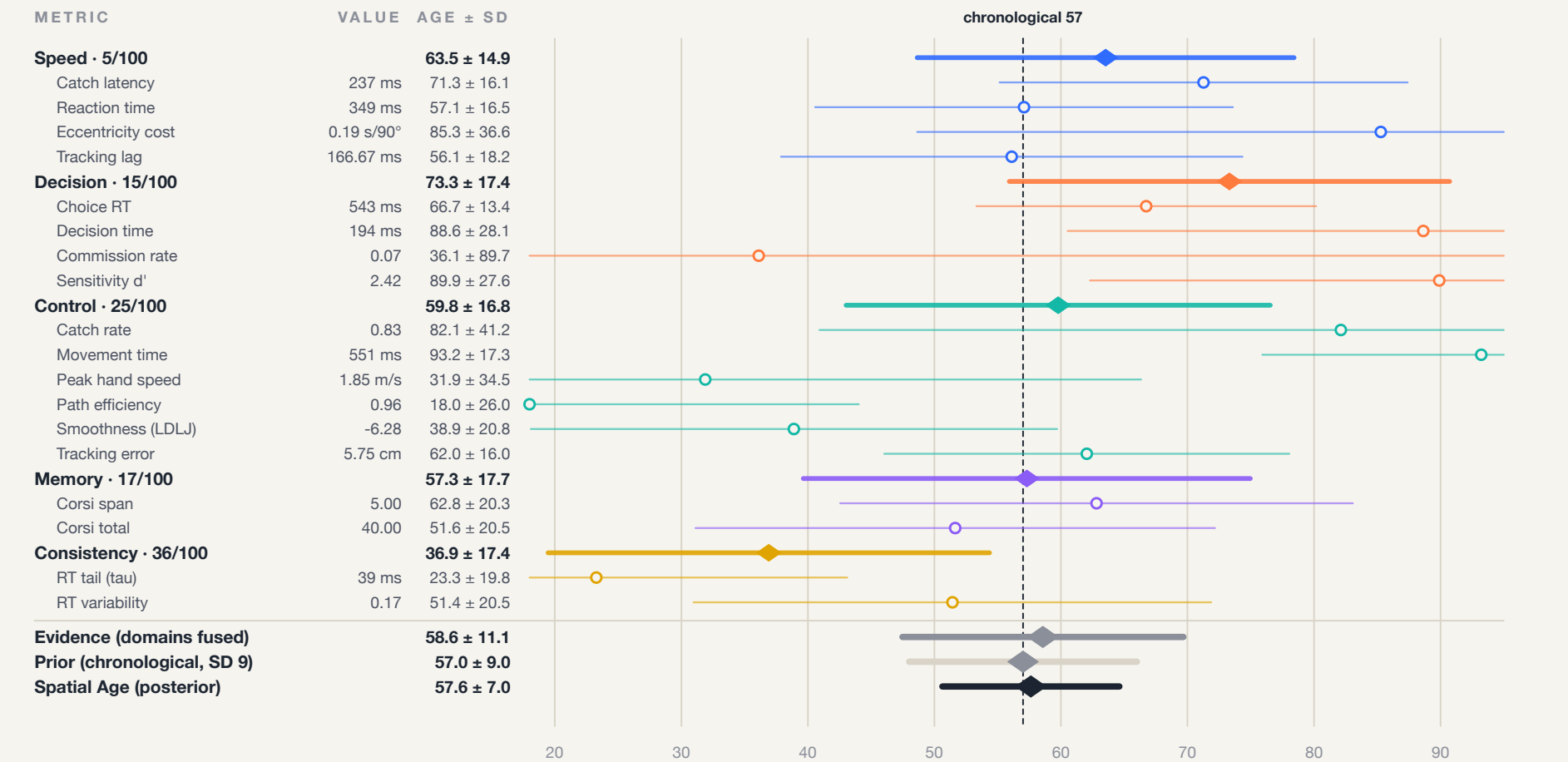
$$\sigma_{\hat{a}} = \sqrt{(SE / |E'(\hat{a})|)^2 + \tau^2}$$

Measurement error converted to years through the local slope, plus the metric's residual error τ . Where the curve is steep a small difference in milliseconds maps to few years, so older ages get tighter estimates.

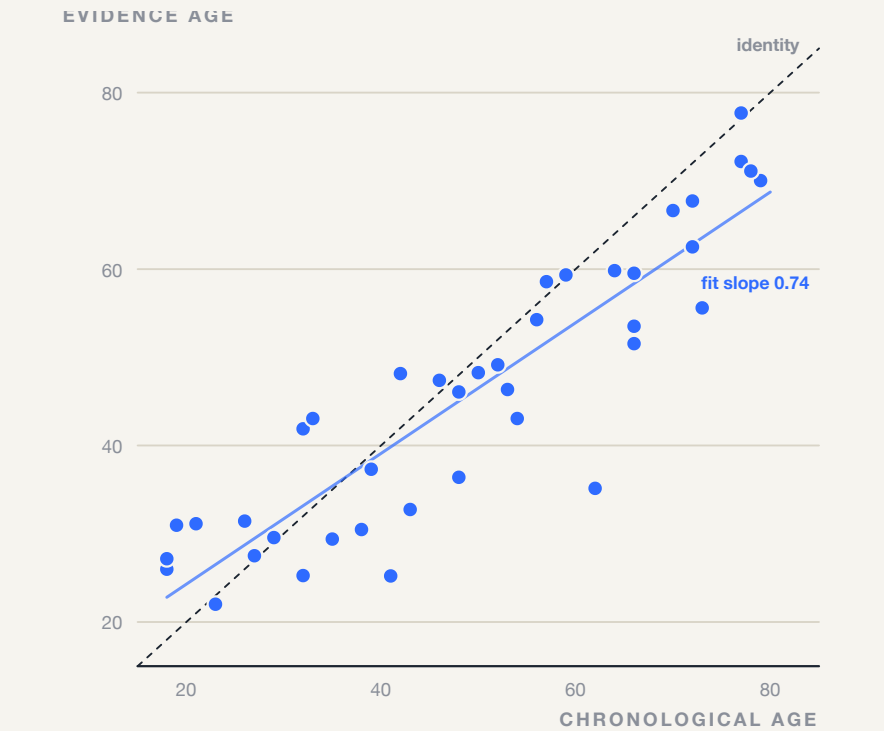
$$\text{reach_rt}: \mu_{25} \ 0.3 \text{ s} \cdot \beta \ 0.0015 \text{ s/y} \cdot \gamma \ 0.00002 \cdot \tau \ 13 \text{ y}$$

Metrics fuse into domains, domains into one age

Inverse-variance mean, variance $\times (1 + (k - 1) \rho)$. $\rho = 0.5$ inside a domain, 0.3 between domains. A chronological prior with SD 9 y pulls thin evidence toward the calendar. Interval is 80%.

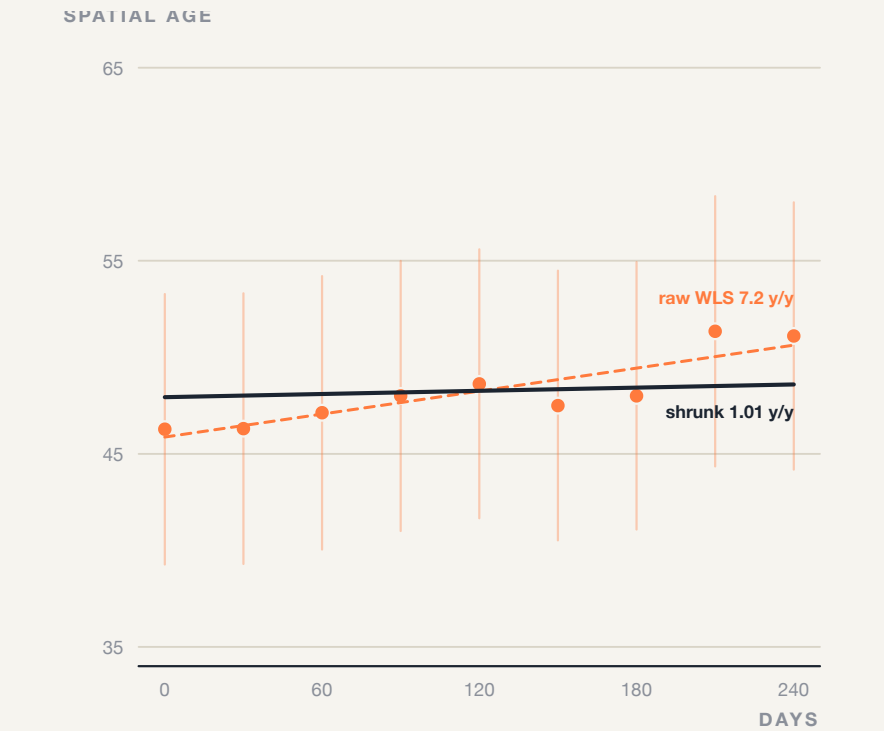


Cohort accuracy and pace of aging



7.0 y **16.1 y** **0.90**
EVIDENCE MAE PREDICT-THE-MEAN PEARSON R

n = 40, ages 18 to 79, 40 usable. Evidence age excludes the chronological prior; the posterior is not an accuracy measure because it already contains the answer. Synthetic biological age = chronological + N(0, 6), so part of the error is real signal. Fit slope 0.74 shows regression toward the cohort middle.



1.01 **± 0.50** **0.80**
PACE, Y PER Y POSTERIOR SD SIMULATED TRUTH

One participant, 9 sessions over 240 days. WLS slope of Spatial Age on time, weights 1/SD², shrunk toward 1.0 with prior SD 0.5. Raw slope 7.2 ± 11 y/y, so the estimate stays at the prior. Session SD needs to fall well under 7 y before pace is measurable inside a year.

Built, in progress, open

BUILT	ScoreKit 0.2.0 Swift package, pure function session → report. 5 extractors, kinematics, robust stats, age model, pace of aging, CLI (score, pace, synth, history, norms). 15 tests pass.
BUILT	Session schema 0.2.0 One JSON contract shared by the app, ingest, ML and dashboard. Report JSON uses snake_case and is versioned by engine and norm table.
IN PROGRESS	Norm table prior-0.2 Curve shapes come from published reaction time, reach and Corsi studies. None of them used a fingertip reach in a headset, so every μ , β , γ and τ is provisional until refit.
IN PROGRESS	Calibration of uncertainty Mean evidence SD is 11.1 y, which predicts an MAE near 8.9 y. Observed MAE is 7.0 y, so intervals are wider than the error. The refit should estimate τ from data.
IN PROGRESS	Age model v1 Ridge regression with leave-one-out CV on event sessions, reported against the predict-the-mean baseline, with linear bias correction for the age gap.
OPEN	Assumed constants $\rho = 0.5 / 0.3$ are assumed, not estimated. τ -only metrics without a trial SE (rt_tau, rt_cv, d', corsi_total) default to $SD_{25} / 3$. Corsi span SE is fixed at 0.5. Constellation does not yet apply the tracking gap filter.
OPEN	Reliability Test-retest on 5 participants, practice effect size across repeat sessions, and the session count needed for a usable pace estimate.

Not a clinical or diagnostic tool. Raw gaze is not used; visionOS does not expose it to apps.